Ringtest auf verschiedenen Rechnerchips – Messbericht

Lauf-Nummer 42529106396 · Stand der Programme: 5fb5bfd94229

1. Ergebnis

27 Rechner liefen auf 9 verschiedenen Prozessortypen. 27 von 27 Läufen bestanden alle fünf Prüfungen.

Auf allen Rechnern ist der exakte Ring (nur Addition, Subtraktion, Multiplikation) Bit für Bit gleich: 1 verschiedene Prüfwert(e) über alle Schritte. Der Phasenlauf mit Sinus liegt auf allen Rechnern innerhalb von 1e-11 zur Referenz: 3 verschiedene Prüfwert(e) des Endzustands.

2. Prüfungen

T1 Ring mit 36 Knoten, 400 Schritte: Prüfwert aller Schritte und des Endzustands gleich der Festlegung, Summe erhalten (< 1e-12), Abfall der langsamsten Schwingung auf 1e-9.

T2 Phasenlauf, 36 Knoten, 400 Schritte: Windung 1 zu allen 401 Zeitpunkten (< 1e-9), jeder Knoten innerhalb 1e-11 zur Referenz, Nachbardifferenzen in [0,13; 0,21], Mittelwert-Drift < 1e-12.

T3 Ring mit 36 Knoten, ε = 0,496, 200 Schritte: Abfall der langsamsten Schwingung 0,984929 und der schnellsten 0,984, Abweichung < 1e-9.

T4 Ringe mit 30 und 70 Knoten: Summe der Eigenwerte 2N, Produkt N^2, Summe der Kehrwerte $(N^2-1)/12$ und Abfall der langsamsten Schwingung, Abweichung < 1e-9.

T5 Fingerabdruck der Rechenbibliothek: 14 000 Werte von Sinus, Kosinus, Exponential, Logarithmus, Potenz, Arkustangens, Tangens; Prüfwert wird je Rechner aufgezeichnet.

3. Prozessortypen

Prozessor	Anzahl Läufe
AMD EPYC 7763 64-Core Processor	4
AMD EPYC 9V45 96-Core Processor	5
AMD EPYC 9V74 80-Core Processor	2
ARM 0x41 / 0xd49	3
Apple M1 (Virtual)	3
Cobalt 100	3
INTEL(R) XEON(R) PLATINUM 8573C	1
Intel(R) Core(TM) i7-8700B CPU @ 3.20GHz	3
Neoverse-N2	3

Fingerabdrücke der Rechenbibliothek (T5): 6 verschiedene. Gleiche Fingerabdrücke bedeuten gleiche Rechenbibliothek, nicht gleichen Prozessor.

4. Alle Läufe

Rechner	System	Architektur	Prozessor	T1	T2	T3	T4	Ergebnis
macos-15-intel_1	Darwin	x86_64	Intel(R) Core(TM) i7-8700B CPU @ 3.20GHz	bestanden	bestanden	bestanden	bestanden	bestanden
macos-15-intel_2	Darwin	x86_64	Intel(R) Core(TM) i7-8700B CPU @ 3.20GHz	bestanden	bestanden	bestanden	bestanden	bestanden
macos-15-intel_3	Darwin	x86_64	Intel(R) Core(TM) i7-8700B CPU @ 3.20GHz	bestanden	bestanden	bestanden	bestanden	bestanden
macos-15_1	Darwin	arm64	Apple M1 (Virtual)	bestanden	bestanden	bestanden	bestanden	bestanden
macos-15_2	Darwin	arm64	Apple M1 (Virtual)	bestanden	bestanden	bestanden	bestanden	bestanden
macos-15_3	Darwin	arm64	Apple M1 (Virtual)	bestanden	bestanden	bestanden	bestanden	bestanden
ubuntu-22.04-arm_1	Linux	aarch64	ARM 0x41 / 0xd49	bestanden	bestanden	bestanden	bestanden	bestanden
ubuntu-22.04-arm_2	Linux	aarch64	ARM 0x41 / 0xd49	bestanden	bestanden	bestanden	bestanden	bestanden
ubuntu-22.04-arm_3	Linux	aarch64	ARM 0x41 / 0xd49	bestanden	bestanden	bestanden	bestanden	bestanden
ubuntu-22.04_1	Linux	x86_64	AMD EPYC 7763 64-Core Processor	bestanden	bestanden	bestanden	bestanden	bestanden
ubuntu-22.04_2	Linux	x86_64	AMD EPYC 7763 64-Core Processor	bestanden	bestanden	bestanden	bestanden	bestanden
ubuntu-22.04_3	Linux	x86_64	AMD EPYC 9V45 96-Core Processor	bestanden	bestanden	bestanden	bestanden	bestanden
ubuntu-24.04-arm_1	Linux	aarch64	Neoverse-N2	bestanden	bestanden	bestanden	bestanden	bestanden
ubuntu-24.04-arm_2	Linux	aarch64	Neoverse-N2	bestanden	bestanden	bestanden	bestanden	bestanden
ubuntu-24.04-arm_3	Linux	aarch64	Neoverse-N2	bestanden	bestanden	bestanden	bestanden	bestanden
ubuntu-24.04_1	Linux	x86_64	AMD EPYC 7763 64-Core Processor	bestanden	bestanden	bestanden	bestanden	bestanden
ubuntu-24.04_2	Linux	x86_64	AMD EPYC 9V45 96-Core Processor	bestanden	bestanden	bestanden	bestanden	bestanden
ubuntu-24.04_3	Linux	x86_64	AMD EPYC 9V74 80-Core Processor	bestanden	bestanden	bestanden	bestanden	bestanden
windows-11-arm_1	Windows	ARM64	Cobalt 100	bestanden	bestanden	bestanden	bestanden	bestanden
windows-11-arm_2	Windows	ARM64	Cobalt 100	bestanden	bestanden	bestanden	bestanden	bestanden
windows-11-arm_3	Windows	ARM64	Cobalt 100	bestanden	bestanden	bestanden	bestanden	bestanden
windows-2022_1	Windows	AMD64	AMD EPYC 9V45 96-Core Processor	bestanden	bestanden	bestanden	bestanden	bestanden
windows-2022_2	Windows	AMD64	AMD EPYC 9V45 96-Core Processor	bestanden	bestanden	bestanden	bestanden	bestanden
windows-2022_3	Windows	AMD64	INTEL(R) XEON(R) PLATINUM 8573C	bestanden	bestanden	bestanden	bestanden	bestanden
windows-2025_1	Windows	AMD64	AMD EPYC 7763 64-Core Processor	bestanden	bestanden	bestanden	bestanden	bestanden
windows-2025_2	Windows	AMD64	AMD EPYC 9V74 80-Core Processor	bestanden	bestanden	bestanden	bestanden	bestanden
windows-2025_3	Windows	AMD64	AMD EPYC 9V45 96-Core Processor	bestanden	bestanden	bestanden	bestanden	bestanden

5. Messwerte je Lauf

Rechner	T1 Abfall (gemessen)	Windung max. Abw.	Knoten max. Abw.	T3 langsam	T4 N=30	T4 N=70
macos-15-intel_1	0.992403876506104	4.44e-16	1.39e-17	0.984929290988110	4.29e-15	1.67e-14
macos-15-intel_2	0.992403876506104	4.44e-16	1.39e-17	0.984929290988110	4.29e-15	1.67e-14
macos-15-intel_3	0.992403876506104	4.44e-16	1.39e-17	0.984929290988110	4.29e-15	1.67e-14
macos-15_1	0.992403876506104	4.44e-16	1.39e-17	0.984929290988110	2.53e-15	1.67e-14
macos-15_2	0.992403876506104	4.44e-16	1.39e-17	0.984929290988110	2.53e-15	1.67e-14
macos-15_3	0.992403876506104	4.44e-16	1.39e-17	0.984929290988110	2.53e-15	1.67e-14
ubuntu-22.04-arm_1	0.992403876506104	4.44e-16	0.00e+00	0.984929290988110	4.29e-15	1.67e-14
ubuntu-22.04-arm_2	0.992403876506104	4.44e-16	0.00e+00	0.984929290988110	4.29e-15	1.67e-14
ubuntu-22.04-arm_3	0.992403876506104	4.44e-16	0.00e+00	0.984929290988110	4.29e-15	1.67e-14
ubuntu-22.04_1	0.992403876506104	4.44e-16	0.00e+00	0.984929290988110	4.29e-15	1.67e-14
ubuntu-22.04_2	0.992403876506104	4.44e-16	0.00e+00	0.984929290988110	4.29e-15	1.67e-14
ubuntu-22.04_3	0.992403876506104	4.44e-16	0.00e+00	0.984929290988110	4.29e-15	1.67e-14
ubuntu-24.04-arm_1	0.992403876506104	4.44e-16	0.00e+00	0.984929290988110	4.29e-15	1.67e-14
ubuntu-24.04-arm_2	0.992403876506104	4.44e-16	0.00e+00	0.984929290988110	4.29e-15	1.67e-14
ubuntu-24.04-arm_3	0.992403876506104	4.44e-16	0.00e+00	0.984929290988110	4.29e-15	1.67e-14
ubuntu-24.04_1	0.992403876506104	4.44e-16	0.00e+00	0.984929290988110	4.29e-15	1.67e-14
ubuntu-24.04_2	0.992403876506104	4.44e-16	0.00e+00	0.984929290988110	4.29e-15	1.67e-14
ubuntu-24.04_3	0.992403876506104	4.44e-16	0.00e+00	0.984929290988110	4.29e-15	1.67e-14
windows-11-arm_1	0.992403876506104	4.44e-16	1.39e-17	0.984929290988110	4.29e-15	1.67e-14
windows-11-arm_2	0.992403876506104	4.44e-16	1.39e-17	0.984929290988110	4.29e-15	1.67e-14
windows-11-arm_3	0.992403876506104	4.44e-16	1.39e-17	0.984929290988110	4.29e-15	1.67e-14
windows-2022_1	0.992403876506104	4.44e-16	1.39e-17	0.984929290988110	4.29e-15	1.67e-14
windows-2022_2	0.992403876506104	4.44e-16	1.39e-17	0.984929290988110	4.29e-15	1.67e-14
windows-2022_3	0.992403876506104	4.44e-16	1.39e-17	0.984929290988110	4.29e-15	1.67e-14
windows-2025_1	0.992403876506104	4.44e-16	1.39e-17	0.984929290988110	4.29e-15	1.67e-14
windows-2025_2	0.992403876506104	4.44e-16	1.39e-17	0.984929290988110	4.29e-15	1.67e-14
windows-2025_3	0.992403876506104	4.44e-16	1.39e-17	0.984929290988110	4.29e-15	1.67e-14

Prüfsumme der Gesamtdaten (ergebnisse/gesamt.json):

`0bf1299a067abc3f73653b05e88777d67342a38d7d3bc3b166f84161f5328350`

Ring test on different computer chips – measurement report

Run number 42529106396 · program version: 5fb5bfd94229

1. Result

27 machines ran on 9 different processor types. 27 of 27 runs passed all five checks.

On all machines the exact ring (only addition, subtraction, multiplication) is bit for bit identical: 1 distinct checksum(s) over all steps. The phase run with sine lies within 1e-11 of the reference on all machines: 3 distinct checksum(s) of the final state.

2. Checks

T1 ring with 36 nodes, 400 steps: checksum of all steps and of the final state equal to the fixed value, sum conserved (< 1e-12), decay of the slowest oscillation to 1e-9.

T2 phase run, 36 nodes, 400 steps: winding 1 at all 401 times (< 1e-9), every node within 1e-11 of the reference, neighbour differences in [0.13; 0.21], mean drift < 1e-12.

T3 ring with 36 nodes, ε = 0.496, 200 steps: decay of the slowest oscillation 0.984929 and of the fastest 0.984, deviation < 1e-9.

T4 rings with 30 and 70 nodes: sum of eigenvalues 2N, product N^2, sum of reciprocals $(N^2–1)/12$ and decay of the slowest oscillation, deviation < 1e-9.

T5 fingerprint of the math library: 14,000 values of sine, cosine, exponential, logarithm, power, arctangent, tangent; the checksum is recorded per machine.

3. Processor types

Processor	Number of runs
AMD EPYC 7763 64-Core Processor	4
AMD EPYC 9V45 96-Core Processor	5
AMD EPYC 9V74 80-Core Processor	2
ARM 0x41 / 0xd49	3
Apple M1 (Virtual)	3
Cobalt 100	3
INTEL(R) XEON(R) PLATINUM 8573C	1
Intel(R) Core(TM) i7-8700B CPU @ 3.20GHz	3
Neoverse-N2	3

Math library fingerprints (T5): 6 distinct. Equal fingerprints mean the same math library, not the same processor.

4. All runs

Machine	System	Architecture	Processor	T1	T2	T3	T4	Result
macos-15-intel_1	Darwin	x86_64	Intel(R) Core(TM) i7-8700B CPU @ 3.20GHz	passed	passed	passed	passed	passed
macos-15-intel_2	Darwin	x86_64	Intel(R) Core(TM) i7-8700B CPU @ 3.20GHz	passed	passed	passed	passed	passed
macos-15-intel_3	Darwin	x86_64	Intel(R) Core(TM) i7-8700B CPU @ 3.20GHz	passed	passed	passed	passed	passed
macos-15_1	Darwin	arm64	Apple M1 (Virtual)	passed	passed	passed	passed	passed
macos-15_2	Darwin	arm64	Apple M1 (Virtual)	passed	passed	passed	passed	passed
macos-15_3	Darwin	arm64	Apple M1 (Virtual)	passed	passed	passed	passed	passed
ubuntu-22.04-arm_1	Linux	aarch64	ARM 0x41 / 0xd49	passed	passed	passed	passed	passed
ubuntu-22.04-arm_2	Linux	aarch64	ARM 0x41 / 0xd49	passed	passed	passed	passed	passed
ubuntu-22.04-arm_3	Linux	aarch64	ARM 0x41 / 0xd49	passed	passed	passed	passed	passed
ubuntu-22.04_1	Linux	x86_64	AMD EPYC 7763 64-Core Processor	passed	passed	passed	passed	passed
ubuntu-22.04_2	Linux	x86_64	AMD EPYC 7763 64-Core Processor	passed	passed	passed	passed	passed
ubuntu-22.04_3	Linux	x86_64	AMD EPYC 9V45 96-Core Processor	passed	passed	passed	passed	passed
ubuntu-24.04-arm_1	Linux	aarch64	Neoverse-N2	passed	passed	passed	passed	passed
ubuntu-24.04-arm_2	Linux	aarch64	Neoverse-N2	passed	passed	passed	passed	passed
ubuntu-24.04-arm_3	Linux	aarch64	Neoverse-N2	passed	passed	passed	passed	passed
ubuntu-24.04_1	Linux	x86_64	AMD EPYC 7763 64-Core Processor	passed	passed	passed	passed	passed
ubuntu-24.04_2	Linux	x86_64	AMD EPYC 9V45 96-Core Processor	passed	passed	passed	passed	passed
ubuntu-24.04_3	Linux	x86_64	AMD EPYC 9V74 80-Core Processor	passed	passed	passed	passed	passed
windows-11-arm_1	Windows	ARM64	Cobalt 100	passed	passed	passed	passed	passed
windows-11-arm_2	Windows	ARM64	Cobalt 100	passed	passed	passed	passed	passed
windows-11-arm_3	Windows	ARM64	Cobalt 100	passed	passed	passed	passed	passed
windows-2022_1	Windows	AMD64	AMD EPYC 9V45 96-Core Processor	passed	passed	passed	passed	passed
windows-2022_2	Windows	AMD64	AMD EPYC 9V45 96-Core Processor	passed	passed	passed	passed	passed
windows-2022_3	Windows	AMD64	INTEL(R) XEON(R) PLATINUM 8573C	passed	passed	passed	passed	passed
windows-2025_1	Windows	AMD64	AMD EPYC 7763 64-Core Processor	passed	passed	passed	passed	passed
windows-2025_2	Windows	AMD64	AMD EPYC 9V74 80-Core Processor	passed	passed	passed	passed	passed
windows-2025_3	Windows	AMD64	AMD EPYC 9V45 96-Core Processor	passed	passed	passed	passed	passed

5. Measured values per run

Machine	T1 decay (measured)	winding max dev.	node max dev.	T3 slow	T4 N=30	T4 N=70
macos-15-intel_1	0.992403876506104	4.44e-16	1.39e-17	0.984929290988110	4.29e-15	1.67e-14
macos-15-intel_2	0.992403876506104	4.44e-16	1.39e-17	0.984929290988110	4.29e-15	1.67e-14
macos-15-intel_3	0.992403876506104	4.44e-16	1.39e-17	0.984929290988110	4.29e-15	1.67e-14
macos-15_1	0.992403876506104	4.44e-16	1.39e-17	0.984929290988110	2.53e-15	1.67e-14
macos-15_2	0.992403876506104	4.44e-16	1.39e-17	0.984929290988110	2.53e-15	1.67e-14
macos-15_3	0.992403876506104	4.44e-16	1.39e-17	0.984929290988110	2.53e-15	1.67e-14
ubuntu-22.04-arm_1	0.992403876506104	4.44e-16	0.00e+00	0.984929290988110	4.29e-15	1.67e-14
ubuntu-22.04-arm_2	0.992403876506104	4.44e-16	0.00e+00	0.984929290988110	4.29e-15	1.67e-14
ubuntu-22.04-arm_3	0.992403876506104	4.44e-16	0.00e+00	0.984929290988110	4.29e-15	1.67e-14
ubuntu-22.04_1	0.992403876506104	4.44e-16	0.00e+00	0.984929290988110	4.29e-15	1.67e-14
ubuntu-22.04_2	0.992403876506104	4.44e-16	0.00e+00	0.984929290988110	4.29e-15	1.67e-14
ubuntu-22.04_3	0.992403876506104	4.44e-16	0.00e+00	0.984929290988110	4.29e-15	1.67e-14
ubuntu-24.04-arm_1	0.992403876506104	4.44e-16	0.00e+00	0.984929290988110	4.29e-15	1.67e-14
ubuntu-24.04-arm_2	0.992403876506104	4.44e-16	0.00e+00	0.984929290988110	4.29e-15	1.67e-14
ubuntu-24.04-arm_3	0.992403876506104	4.44e-16	0.00e+00	0.984929290988110	4.29e-15	1.67e-14
ubuntu-24.04_1	0.992403876506104	4.44e-16	0.00e+00	0.984929290988110	4.29e-15	1.67e-14
ubuntu-24.04_2	0.992403876506104	4.44e-16	0.00e+00	0.984929290988110	4.29e-15	1.67e-14
ubuntu-24.04_3	0.992403876506104	4.44e-16	0.00e+00	0.984929290988110	4.29e-15	1.67e-14
windows-11-arm_1	0.992403876506104	4.44e-16	1.39e-17	0.984929290988110	4.29e-15	1.67e-14
windows-11-arm_2	0.992403876506104	4.44e-16	1.39e-17	0.984929290988110	4.29e-15	1.67e-14
windows-11-arm_3	0.992403876506104	4.44e-16	1.39e-17	0.984929290988110	4.29e-15	1.67e-14
windows-2022_1	0.992403876506104	4.44e-16	1.39e-17	0.984929290988110	4.29e-15	1.67e-14
windows-2022_2	0.992403876506104	4.44e-16	1.39e-17	0.984929290988110	4.29e-15	1.67e-14
windows-2022_3	0.992403876506104	4.44e-16	1.39e-17	0.984929290988110	4.29e-15	1.67e-14
windows-2025_1	0.992403876506104	4.44e-16	1.39e-17	0.984929290988110	4.29e-15	1.67e-14
windows-2025_2	0.992403876506104	4.44e-16	1.39e-17	0.984929290988110	4.29e-15	1.67e-14
windows-2025_3	0.992403876506104	4.44e-16	1.39e-17	0.984929290988110	4.29e-15	1.67e-14

Checksum of the complete data (ergebnisse/gesamt.json):

`0bf1299a067abc3f73653b05e88777d67342a38d7d3bc3b166f84161f5328350`